

Devansh Vashishat

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EDUCATION

Thapar Institute of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.0/10.0

Patiala, Punjab

Aug. 2023 – June 2027

EXPERIENCE

Software Development Engineer Intern

June 2025 – Dec. 2025

Recruitment and Assessment Centre (RAC), DRDO

New Delhi

- Developed AI-driven expert recommendation system using Jaccard similarity, Cosine similarity, and TF-IDF algorithms to automate interview panel formation, reducing manual expert selection time from 8-9 hours to 15 minutes per request.
- Built data ingestion pipeline scraping IRINS pages of universities to extract professor expertise and publications, expanding searchable expert database beyond internal DRDO employees to include academics from reputed institutions.
- Designed and deployed backend using Node.js and Express.js with PostgreSQL database (migrated from MongoDB prototype), handling role as Backend Developer and DevOps engineer for internal RAC tool.

Research Intern – Samsung PRISM Program

Jan. 2025 – June 2025

Samsung Pvt Ltd, India

Bengaluru

- Engineered AI-powered worklet generation system using open-source LLMs (Gemma 3 14B, GPT-OSS 20B) with RAG architecture to autonomously create research project documents, reducing worklet creation time from 3-4 hours to 5-10 minutes.
- Developed end-to-end data pipeline integrating Tesseract OCR for document processing and open-source Google Scholar scraper for ingesting historical research data, increasing researcher productivity by 80% based on user surveys.
- Optimized information retrieval performance by 45% through asynchronous functions and parallel processing architecture, validated through user feedback and time-tracking surveys.
- Built full-stack web interface for researchers with streamlined data ingestion workflow, enabling automated processing of research documents and scholarly publications.

Backend & DevOps Intern

Dec. 2024 – Feb. 2025

LMTSM

Patiala, Punjab

- Developed student startup listing platform serving 30+ ventures with fuzzy search functionality and optimized PostgreSQL database indexing, improving product discoverability and reach.
- Implemented CI/CD pipeline using GitHub Actions to automate deployment workflow, eliminating manual server login and deployment steps, enabling continuous deployment with single push to repository.

PROJECTS

AI-Driven Expert Recommendation Engine – SIH 2024 Winner

- Built scalable recommendation system for DRDO to automate expert selection and interview panel formation using optimized RESTful APIs and real-time data pipelines, eliminating hours of manual effort per request.
- Implemented automated data extraction from university IRINS portals and integrated intelligent matching algorithms (Jaccard, Cosine, TF-IDF) to expand expert coverage and improve selection accuracy across diverse technical domains.
- Developed complete backend architecture using Node.js, Express.js, and PostgreSQL with DevOps automation, successfully transitioning from hackathon prototype to production-ready internal tool for RAC, DRDO.

Privacy-First Document Query Engine | [Live Demo](#) | *LangGraph, LLMs, ChromaDB, MongoDB*

- Built local-first agentic RAG application using LangGraph for multi-format document processing (PDF, DOCX, TXT) with semantic chunking and vector search via ChromaDB, ensuring sensitive data remains on user infrastructure.
- Integrated query optimization pipeline with rephraser LLM and retrieval-augmented generation using locally deployed open-source models (Gemma 3 14B, GPT-OSS) on H100 GPUs, enabling complete answers without cloud dependencies.
- Designed hybrid storage architecture using ChromaDB for vector embeddings and MongoDB for user authentication and metadata, optimizing for local deployment with storage and hardware constraints as primary limitations.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL

Backend Development: Node.js, Express.js, FastAPI, RESTful APIs

Machine Learning & AI: LangChain, LangGraph, RAG Architecture, Vector Databases(Chroma DB), NLP

Databases & Storage: PostgreSQL, MongoDB, ChromaDB

DevOps & Tools: Docker, Git, GitHub Actions, CI/CD Pipelines, Linux/Unix, Nginx

Cloud & Infrastructure: AWS (EC2, S3), DigitalOcean, GPU Infrastructure (H100)

ACHIEVEMENTS

- Winner of Smart India Hackathon 2024 among 400+ competing teams for developing AI-driven expert recommendation engine; solution adopted by DRDO and converted into 6-month SDE internship at RAC, DRDO.